

users(user_id, first_name, last_name, email, password_hash, phone, date_of_birth, gender, join_date, is_active)

user_profiles(profile_id, *user_id* → *users.user_id*, height_cm, weight_kg, fitness_level, fitness_goals, medical_conditions, profile_image, bio, updated_at)

trainers(trainer_id, first_name, last_name, email, phone, certification, specialization, bio, profile_image, years_experience, hourly_rate, is_active)

class_schedules(class_id, class_name, description, *trainer_id* → *trainers.trainer_id*, class_date, start_time, end_time, max_participants, location, difficulty_level)

class_enrollments(enrollment_id, *class_id* → *class_schedules.class_id*, *user_id* → *users.user_id*, enrollment_date, attended)

workout_programs(program_id, program_name, description, difficulty_level, duration_weeks, *trainer_id* → *trainers.trainer_id*, created_date, program_image, is_active)

program_exercises(program_exercise_id, *program_id* → workout_programs.program_id, exercise_id → exercises.exercise_id, day_number, sets, reps, rest_seconds, order_in_workout)

exercises(exercise_id, exercise_name, description, muscle_group, equipment_needed, difficulty_level, video_url, image_url, instructions)

user_workouts(workout_id, *user_id* → *users.user_id*, *program_id* → *workout_programs.program_id*, workout_date, duration_minutes, calories_burned, notes, completed, created_at)

workout_exercises(workout_exercise_id, *workout_id* → *user_workouts.workout_id*, exercise_id → *exercises.exericse_id*, sets_completed, reps_completed, weight_used, notes)

goals(goal_id, *user_id* → *users.user_id*, goal_type, target_value, current_value, target_date, achieved, created_date, notes)

nutrition_logs(log_id, *user_id* → *users.user_id*, log_date, meal_type, meal_name, calories, protein_g, carbs_g, fat_g, notes, created_at)

progress_tracking(progress_id, *user_id* → *users.user_id*, measurement_date, weight_kg, body_fat_percentage, muscle_mass_kg, chest_cm, waist_cm, hips_cm, biceps_cm, thighs_cm, notes)

USERS(user_id, first_name, last_name, email, password_hash, phone, date_of_birth, gender, join_date, is_active)

PK: user_id

Candidate Keys: email

Functional Dependencies

user_id → first_name, last_name, email, password_hash, phone, date_of_birth, gender, join_date, is_active

email → user_id

3NF Justification

All non-key attributes depend on the primary key (user_id).

No partial or transitive dependencies.

→ USERS is in 3NF.

user_id	first_name	last_name	email	is_active
1	Taylor	Reed	treed@example.com	1
2	Jordan	Smith	jsmith@example.com	1
3	Ava	Carter	acarter@example.com	0

USER_PROFILES(profile_id, user_id, height_cm, weight_kg, fitness_level, fitness_goals, medical_conditions, profile_image, bio, updated_at)

PK: profile_id

FK: user_id → USERS.user_id

Functional Dependencies

profile_id → user_id, height_cm, weight_kg, fitness_level, fitness_goals, medical_conditions, profile_image, bio, updated_at

3NF Justification

All attributes depend on profile_id.

No transitive dependencies.

→ USER_PROFILES is in 3NF.

profile_id	user_id	height_cm	weight_kg	fitness_level
1	1	170	68	Intermediate
2	2	178	74	Beginner
3	3	165	60	Advanced

TRAINERS(trainer_id, first_name, last_name, email, phone, certification, specialization, bio, profile_image, years_experience, hourly_rate, is_active)

PK: trainer_id

Candidate Keys: email

Functional Dependencies

trainer_id → all attributes

email → trainer_id

3NF Justification

All non-key attributes depend directly on trainer_id.

→ TRAINERS is in 3NF.

trainer_id	first_name	specialization	hourly_rate
1	Mia	Strength	45
2	Leo	Cardio	40
3	Sam	HIIT	50

CLASS_SCHEDULES(class_id, class_name, description, trainer_id, class_date, start_time, end_time, max_participants, location, difficulty_level)

PK: class_id

FK: trainer_id → TRAINERS.trainer_id

Functional Dependencies

class_id → all attributes

3NF Justification

Attributes depend only on class_id.

→ CLASS_SCHEDULES is in 3NF.

class_id	class_name	trainer_id	location
1	Morning Yoga	2	Studio A
2	HIIT Blast	3	Gym Room
3	Strength 101	1	Studio B

CLASS_ENROLLMENTS(enrollment_id, class_id, user_id, enrollment_date, attended)

PK: enrollment_id

FKs:

class_id → CLASS_SCHEDULES.class_id

user_id → USERS.user_id

Functional Dependencies

enrollment_id → class_id, user_id, enrollment_date, attended

3NF Justification

All attributes depend on enrollment_id.

→ CLASS_ENROLLMENTS is in 3NF.

enrollment_id	class_id	user_id	attended
1	1	2	1
2	2	1	0
3	3	3	1

WORKOUT_PROGRAMS(program_id, program_name, description, difficulty_level, duration_weeks, trainer_id, created_date, program_image, is_active)

PK: program_id

FK: trainer_id → TRAINERS.trainer_id

Functional Dependencies

program_id → all attributes

3NF Justification

Fully functionally dependent on program_id.

→ WORKOUT_PROGRAMS is in 3NF.

program_id	program_name	trainer_id	duration_weeks
1	Beginner Strength	1	6
2	Fat Burn HIIT	3	4
3	Yoga Flow	2	8

PROGRAM_EXERCISES(program_exercise_id, program_id, exercise_id, day_number, sets, reps, rest_seconds, order_in_workout)

PK: program_exercise_id

FK: program_id → WORKOUT_PROGRAMS.program_id

FK: exercise_id → EXERCISES.exercise_id

Functional Dependencies

program_exercise_id → all attributes

3NF Justification

No transitive dependencies.

→ PROGRAM_EXERCISES is in 3NF.

program_exercise_id	program_id	exercise_id	sets
1	1	5	4
2	1	2	3
3	2	1	5

EXERCISES(exercise_id, exercise_name, description, muscle_group, equipment_needed, difficulty_level, video_url, image_url, instructions)

PK: exercise_id

Functional Dependencies

exercise_id → all attributes

3NF Justification

All attributes depend on exercise_id.

→ EXERCISES is in 3NF.

exercise_id	exercise_name	muscle_group
1	Push-Up	Chest
2	Squat	Legs
3	Plank	Core

USER_WORKOUTS(workout_id, user_id, program_id, workout_date, duration_minutes, calories_burned, notes, completed, created_at)

PK: workout_id

FK: user_id → USERS.user_id

FK: program_id → WORKOUT_PROGRAMS.program_id

Functional Dependencies

workout_id → all attributes

3NF Justification

All attributes depend on workout_id.

→ USER_WORKOUTS is in 3NF.

workout_id	user_id	program_id	completed
1	1	1	1
2	2	3	0
3	3	2	1

WORKOUT_EXERCISES(workout_exercise_id, workout_id, exercise_id, sets_completed, reps_completed, weight_used, notes)

PK: workout_exercise_id

FK: workout_id → USER_WORKOUTS.workout_id

FK: exercise_id → EXERCISES.exercise_id

Functional Dependencies

workout_exercise_id → all attributes

3NF Justification

Attributes depend solely on PK.

→ WORKOUT_EXERCISES is in 3NF.

workout_exercise_id	workout_id	exercise_id	sets_completed
1	1	1	4
2	1	3	1
3	2	2	3

GOALS(goal_id, user_id, goal_type, target_value, current_value, target_date, achieved, created_date, notes)

PK: goal_id

FK: user_id → USERS.user_id

Functional Dependencies

goal_id → all attributes

3NF Justification

Fully dependent on goal_id.

→ GOALS is in 3NF.

goal_id	user_id	goal_type	achieved
1	1	Weight Loss	0
2	2	Muscle Gain	1
3	3	Flexibility	0

NUTRITION_LOGS(log_id, user_id, log_date, meal_type, meal_name, calories, protein_g, carbs_g, fat_g, notes, created_at)

PK: log_id

FK: user_id → USERS.user_id

Functional Dependencies

log_id → all attributes

3NF Justification

Attributes depend on log_id.

→ NUTRITION_LOGS is in 3NF.

log_id	user_id	meal_type	calories
1	1	Breakfast	400
2	1	Lunch	600
3	2	Dinner	550

PROGRESS_TRACKING(progress_id, user_id, measurement_date, weight_kg, body_fat_percentage, muscle_mass_kg, chest_cm, waist_cm, hips_cm, biceps_cm, thighs_cm, notes)

PK: progress_id

FK: user_id → USERS.user_id

Functional Dependencies

progress_id → all attributes

3NF Justification

All attributes depend on progress_id.

→ PROGRESS_TRACKING is in 3NF.

progress_id	user_id	weight_kg	waist_cm
1	1	68	72
2	1	67	71
3	2	74	82